
QBUS2810

Statistical Modelling for Business

Semester 1, 2026

Group Assignment

Daily Bike Sharing Demand: Explanation, Prediction and Model Discipline

Administrative details

Weighting	30% of final mark
Due date	29 May, 2026
Submission	Assignment on Canvas
Recommended report length	18–25 pages, excluding Python code appendix

This assignment must be completed in your Canvas group.

You may use GenAI, but any use must be acknowledged carefully and all prompts used must be included in an appendix. Strong analysis and clear judgement matter more than mechanically produced output.

General Instructions

- You must use Python for this assignment.
- All plots, tables, model output and forecast results discussed in the report must be reproducible from your Python code.
- Include an appendix containing all Python code used to produce the assignment outputs. A heavy penalty applies if the code is missing, does not run, or does not match the submitted report.
- Your final report should be submitted as a professionally edited PDF, not as a raw notebook export.
- There is no strict page limit, but irrelevant, repetitive or poorly edited material will reduce your mark.
- Throughout this assignment, you must distinguish between an explanatory model and an operational prediction model. A modelling choice that is acceptable for one goal may be unsuitable for another.
- For predictive tasks, assume the forecast is made for same-day total demand after calendar information and daily weather expectations are available, but before any user counts for that day are observed.
- Not every column in the data file should automatically be treated as an ordinary predictor. If a variable is contemporaneous with the response, mechanically linked to it, or unavailable at the relevant decision time, then this matters. Ignoring this issue without justification will be penalised.

Pre-analysis Instructions for Data

- Use the file `data/bike_sharing_day_2026.csv` only.
- Open the notebook `generate_train_test_split.ipynb`.
- Replace the value of `state` with the sum of the student ID numbers for the members of your group.
- Run the notebook to create `train` and `test` datasets.
- Use the `train` data for exploratory analysis, estimation, diagnostics, and model selection.
- Use the `test` data only for the final forecast comparison in Task 6.
- If you want an out-of-sample performance measure before Task 6, it must come from a validation approach applied within the training data, such as cross-validation or a training-only validation split.
- The notebook creates a different random split for each group, but it does not tell you which variables are admissible for each modelling goal. Deciding that is part of the assignment.

Business Problem

Capital Bikeshare and transport planners in Washington, D.C. want to better understand the drivers of daily bike rental demand and to build a model that can predict total daily rentals accurately enough to support planning, staffing and bike rebalancing decisions.

Management is also interested in a broader strategic question: should future planning emphasis lean more toward commuter-reliability demand, more toward leisure and fair-weather demand, or remain broadly balanced across both? The data may help answer this, but only if you are careful about the distinction between explanation, prediction and causal interpretation.

The analysts are concerned that some variables in the data file may be unsuitable for one or more modelling goals because they are decompositions of the response, are measured contemporaneously with it, or would not be available in the same way when a forecast is made. Your group has been asked to analyse the data and report back.

Data and Response Variable

The data are daily observations from the Capital Bikeshare system for 2011 and 2012, together with weather and calendar information. The main response variable is the total daily rental count `cnt`, though you may work with a suitable transformation such as `log(cnt)` if well justified.

See `Data_Source_and_Variable_Notes.md` for source details, variable definitions and important cautions.

Primary goals

1. **Explanatory goal:** understand how daily bike demand is associated with weather and calendar conditions, and whether the apparent effect of temperature remains after controlling for important confounders.
2. **Predictive goal:** develop a parsimonious but accurate model for predicting total daily rentals using only variables that are defensible for the prediction setting you choose.
3. **Decision goal:** advise management, in plain English, whether planning emphasis should lean more toward commuter-reliability demand, more toward leisure and fair-weather demand, or remain broadly balanced across both, while being explicit about what the data can and cannot justify as a causal conclusion.

Tasks

Task 1. Exploratory analysis and variable admissibility (5 marks)

Conduct a suitable exploratory analysis relevant to the goals of this study.

Before fitting major models, discuss which variables are:

- clearly admissible,
- admissible only with caveats, or
- inadmissible

for modelling total daily demand. You do not need to present a long table for every variable if a concise discussion is clearer, but you must specifically comment on `casual`, `registered`, and at

least one date or calendar field.

Your exploratory analysis should be purposeful. It should help motivate later modelling choices rather than simply display many unconnected plots.

Task 2. Simple relationship between temperature and total demand (5 marks)

Analyse the relationship between `temp` and `cnt`.

Include:

- suitable plot(s),
- at least one simple model,
- discussion of whether a linear relationship looks plausible,
- discussion of whether the assumptions of your analysis and test could reasonably hold, and
- discussion of which omitted variables are likely to bias the simple relationship and why.

If you transform the response or the predictor, justify that choice.

Task 3. Explanatory multiple regression model (10 marks)

Build an explanatory multiple linear regression model for total daily demand.

Your model must include:

- `temp`,
- at least one calendar or categorical factor, and
- any additional regressors that you judge necessary to address confounding or important structure.

Do not use variables that you have already argued are inadmissible for this goal.

In your discussion:

- test for the existence of a relationship between demand and temperature,
- test for the existence of a relationship between demand and at least one categorical or calendar factor,
- discuss whether the assumptions of your analysis and tests could hold for this data,
- discuss how well the data fit the model, and
- discuss whether multicollinearity is a problem.

Task 4. Predictive model selection under admissibility constraints (15 marks)

In view of Goal 2, conduct a variable and model selection exercise for predicting total daily rentals.

You must compare at least 8 admissible model specifications. Across the set of models you try, include:

- at least 2 interaction effects,
- at least 2 transformations or nonlinear effects on regressors and/or the response, and
- at least one polynomial or spline effect.

You must properly motivate and discuss your modelling choices here. A long list of models with no reasoning will not receive full credit.

Report a concise comparison of the competing models and then select a preferred predictive model using the training data only. If you discuss out-of-sample accuracy at this stage, it should mean cross-validated or validation-sample performance computed within the training data, not performance on the held-out test set. Your chosen model should still be defensible and explainable to management.

For the final preferred model:

- report the model clearly,
- provide suitable diagnostics,
- discuss any collinearity issues, and
- if nonlinear effects are present, clearly interpret them.

Task 5. Outliers, leverage, influence and robustness (5 marks)

Investigate outliers, leverage and influence for at least one strong model fitted on the training data.

Identify at least one influential day and discuss why it may be influential based on the observed variables and diagnostics.

Then perform at least one defensible robustness check. Examples include:

- an alternative specification,
- a transformation,
- a carefully justified exclusion if you believe there is evidence of data error, or
- another well-motivated modelling adjustment.

Discuss whether your substantive conclusions materially change.

Task 6. Forecast evaluation on the test data (10 marks)

Using at least the 3 best admissible predictive models considered so far, generate forecasts for total daily rentals in the `test` dataset.

Present:

- a suitable summary table,
- appropriate plot(s), and
- forecast accuracy measures RMSE, MAD and forecast R^2 .

Re-discuss your conclusions in light of these out-of-sample results.

Do the best explanatory and best predictive models align? Why or why not?

Task 7. Final report for a decision maker (5 marks)

Write a final report, in as close to plain English as practical, for a non-technical decision maker at Capital Bikeshare and/or the Washington, D.C. transport authority.

Your report should:

- summarise the main findings,
- explain the main uncertainties and limitations,
- state clearly which variables should not be treated as ordinary operational predictors for total demand,
- distinguish between empirical association, predictive usefulness and causal interpretation, and
- provide and justify a recommendation about whether planning emphasis should lean more toward commuter-reliability demand, more toward leisure and fair-weather demand, or remain broadly balanced across both.

Notes on Marking

- The assignment will initially be marked out of 55.
- Up to an additional 5 marks will be awarded for the overall quality of presentation.
- Clear, efficient and professional communication will be rewarded.
- Unclear, poorly justified or mechanically generated analysis will lose marks even if the numerical output looks plausible.

Data Source Note

This draft assignment uses the UCI Bike Sharing dataset, daily file only, covering 2011-01-01 to 2012-12-31. The official dataset page is available at <https://archive-beta.ics.uci.edu/dataset/275/bike%2Bsharing%2Bdataset>.